UTS 31005 Machine Learning

Assignment 2 Report

Recall-oriented Brain Tumour Screening from MRI with EfficientNetB0 Transfer Learning

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Model: EfficientNetB0 (ImageNet transfer learning, two-stage fine-tuning)

Dataset: Brain Tumor MRI Dataset (M. Nickparvar, Kaggle) — 7,023 images, 4 classes collapsed to tumour / no-tumour

Run: CPU, TensorFlow 2.21.0, seed 42, 11 epochs, 24.6 min training

Dataset: https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

1. Executive Summary

This project builds a convolutional screening model that flags the presence of a brain tumour in an MRI slice. Because a missed tumour (false negative) is far more harmful than a false alarm in a triage setting, the objective is to **maximise recall** — but under explicit precision and specificity floors so that the model cannot satisfy the objective trivially by labelling every scan as a tumour.

An ImageNet-pretrained EfficientNetB0 was fine-tuned in two stages and a decision threshold $\tau = 0.007$ was selected on the validation set. On the untouched, de-duplicated held-out test set (n = 1284) the model reaches **recall 0.997 (95% CI [0.992, 1.000]), precision 0.949, specificity 0.874, AUC-ROC 0.998 and AUC-PR 0.999**; 3 of 903 tumour scans were missed. Every figure in this report is produced by the accompanying notebook and read from its reports/metrics.json.

2. Introduction & Objective

Radiological screening of MRI volumes is slow and subject to fatigue and inter-reader variability. A high-sensitivity automated first pass can prioritise suspicious studies for a radiologist. The learning problem is a binary image classification task; the deployment goal fixes the criterion: sensitivity first, with enough specificity that the tool actually removes work rather than adding it.

Success criterion. Choose a classifier and operating point that maximise recall subject to precision ≥ 0.9 and specificity ≥ 0.8 on validation data, then report performance once on a held-out test set.

3. Dataset & Pre-processing

- **Source:** Brain Tumor MRI Dataset (Nickparvar, 2021), Kaggle. Folders glioma / meningioma / pituitary / notumor, provided as official Training and Testing splits.
- **Labels:** notumor → 0, any tumour type → 1 (binary screening task). The tumour type is retained as metadata for the error analysis.
- **Data audit:** 7,023 raw files; MD5 hashing found 194 groups of byte-identical images, 79 of which span the official Training and Testing folders. Duplicates were removed (keeping the Testing copy), leaving 6,726 unique images. Without this step a random re-split would leak test images into training and inflate every metric.
- **Split:** official Testing folder = held-out test set (1284); official Training folder split 85/15 into train (4625) / validation (817), stratified by tumour type, seed 42. The test set is read exactly once, in the final evaluation.
- **Pre-processing:** resize to 224×224 (bilinear), raw 0–255 pixels passed to the network — keras.applications.EfficientNet normalises internally, so no manual rescaling is applied (a [0,1] rescale would silently break the pretrained features).
- **Augmentation (training only):** horizontal flip, rotation ±~11°, zoom ±10%, translation ±5%, contrast ±15%, brightness ±10%. Images are cached before the augmentation step so each epoch sees a fresh random draw.
- **Class imbalance:** positive prior 0.752; balanced class weights 2.02 / 0.66 applied to the loss.

Split	No tumour (0)	Tumour (1)	Total	Positive rate
train	1147	3478	4625	0.752
val	203	614	817	0.752
test	381	903	1284	0.703

Table 1 — Class counts after de-duplication.

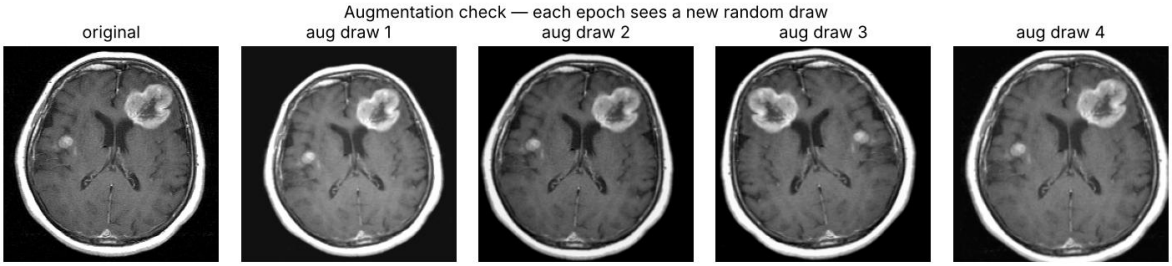


Figure 1 — One training image and four independent augmentation draws.

4. Methodology (DHAL mapping)

DHAL element	This project
Data (D)	6,726 de-duplicated MRI slices with binary labels; official test split held out.
Hypothesis (H)	EfficientNetB0 feature extractor (ImageNet) + global average pooling + dropout 0.3 + sigmoid unit. 4,050,852 parameters.
Algorithm (A)	Adam; stage 1 trains only the head (lr 0.001), stage 2 unfreezes the top 20.0% of backbone layers (38/237, BatchNorm frozen, lr 0.0001); early stopping and LR reduction on validation AUC-PR.
Loss (L)	Class-weighted binary cross-entropy for training; decision threshold τ selected afterwards to maximise recall s.t. precision ≥ 0.9 and specificity ≥ 0.8 (validation only).

Table 2 — DHAL decomposition.

5. Model Architecture & Optimisation

Architecture

- Backbone: EfficientNetB0, ImageNet weights, input 224×224×3, top removed.
- Head: GlobalAveragePooling2D → Dropout(0.3) → Dense(1, sigmoid). The output layer is kept in float32 so mixed precision (GPU runs) cannot saturate the sigmoid.
- Parameters: 4,050,852 total; 1,281 trainable in stage 1, 2,288,785 in stage 2.

Optimisation strategy

1. Stage 1 — frozen backbone, Adam lr 0.001, up to 6 epochs (6 run).
2. Stage 2 — top 20.0% of backbone layers trainable, BatchNormalization layers kept frozen (small batches would corrupt their running statistics), Adam lr 0.0001, up to 5 epochs (5 run).
3. Callbacks: ModelCheckpoint + EarlyStopping (patience 3, best weights restored) + ReduceLROnPlateau (×0.5, patience 2), all monitoring validation AUC-PR. Recall is deliberately not used for model selection: the constant "always tumour" classifier has recall 1.0, so a recall monitor rewards collapse.
4. Batch size 32; balanced class weights; seed 42 for Python, NumPy and TensorFlow.

Training time: 24.6 min on a 2-core CPU; best validation AUC-PR 0.9998 at epoch 10 of 11.

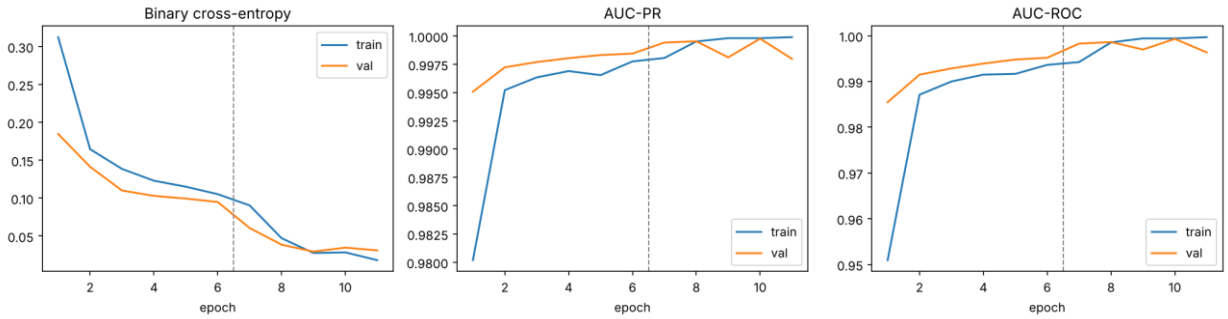


Figure 2 — Learning curves (dashed line = start of fine-tuning). Validation AUC-PR/AUC-ROC saturate quickly; no divergence between train and validation loss is observed.

6. Criterion Alignment — choosing the operating point

The network is trained to minimise cross-entropy, but the deployment criterion is recall under constraints. The two are reconciled after training by sweeping the decision threshold on the validation set. The constraint design matters: the positive prior is about 0.75, so any precision floor below the prior is satisfied at τ = 0 and the "recall-maximising" rule collapses to predicting tumour for every scan. Table 3 shows this on the same validation scores.

Decision rule	τ	Recall	Precision	Specificity	F1	FN	FP
τ=0.50 (default)	0.500	0.987	0.995	0.985	0.991	8	3
recall-max s.t. precision≥0.60 only (original rule)	0.000	1.000	0.751	0.000	0.858	0	203
recall-max s.t. precision≥0.9 & specificity≥0.8 (chosen)	0.007	1.000	0.962	0.882	0.981	0	24

Table 3 — Validation metrics for three decision rules applied to the same model scores.

The rule adopted — maximise recall subject to precision ≥ 0.9 and specificity ≥ 0.8 — gives **τ = 0.007**, validation recall 1.000, precision 0.962, specificity 0.882. Relative to the default τ = 0.5 the number of missed tumours on validation goes from 8 to 0 while false alarms go from 3 to 24; relative to the precision-only rule (τ = 0.000, specificity 0.000) the specificity floor keeps the operating point clinically usable. The floors were not the binding constraint in this run: validation recall reaches 1.0 over a whole range of low thresholds, and the tie-break (highest precision among recall-maximising thresholds) places τ at the top of that plateau — i.e. just below the lowest-scoring validation tumour. The very small τ reflects how well separated the two score distributions are (Figure 8), not a lax criterion.

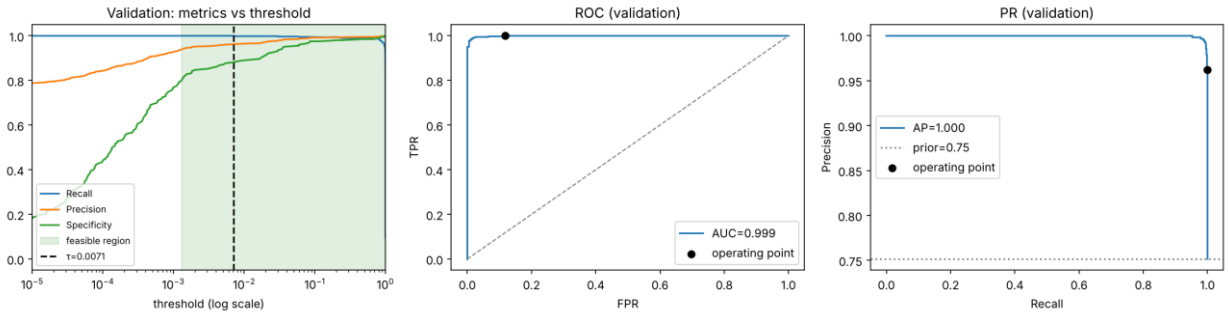


Figure 3 — Validation: recall/precision/specificity vs threshold with the feasible region shaded; ROC and PR curves with the chosen operating point.

7. Evaluation Results

Metric	Validation (threshold selection)	Test (held-out, n = 1284)	Test 95% bootstrap CI
Recall (sensitivity)	1.000	0.997	[0.992, 1.000]
Precision	0.962	0.949	[0.936, 0.963]
Specificity	0.882	0.874	[0.840, 0.907]
F1	0.981	0.972	[0.965, 0.980]
Accuracy	0.971	0.960	—
AUC-ROC	0.999	0.998	[0.996, 0.999]
AUC-PR (AP)	1.000	0.999	[0.999, 1.000]
Brier score	0.010	0.019	—

Table 4 — Metrics at τ = 0.007; the CI is from 1,000 bootstrap resamples of the test set.

Test set	Predicted 0	Predicted 1
True 0 (no tumour)	333	48
True 1 (tumour)	3	900

Table 5 — Test confusion matrix.

Tumour type (test)	n	Correct	Rate	Metric
glioma	299	299	1.000	recall
meningioma	304	301	0.990	recall
pituitary	300	300	1.000	recall
notumor	381	333	0.874	specificity

Table 6 — Per-class detection rate on the test set.

Interpretation. 900 of 903 tumour scans are flagged (recall 0.997); the 48 false alarms among 381 normal scans (specificity 0.874) are the price of the high-sensitivity operating point and are acceptable for a triage tool whose output is reviewed by a radiologist. Test metrics track the validation metrics closely, so the threshold chosen on validation generalises. For comparison, the default cut-off τ = 0.5 would give test recall 0.977 (21 missed tumours) with specificity 0.966 (13 false alarms): the recall-first operating point buys 18 fewer missed tumours at the cost of 35 additional false alarms.

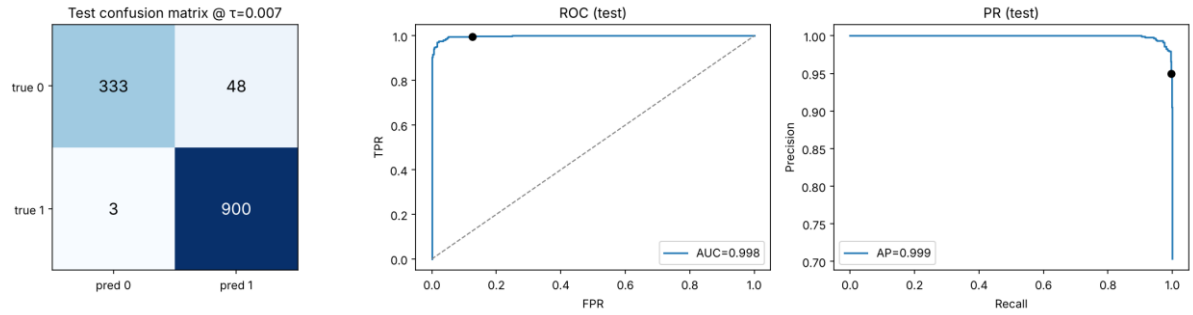


Figure 4 — Test set: confusion matrix, ROC and PR curves with the operating point.

8. Visualisation & Explainability

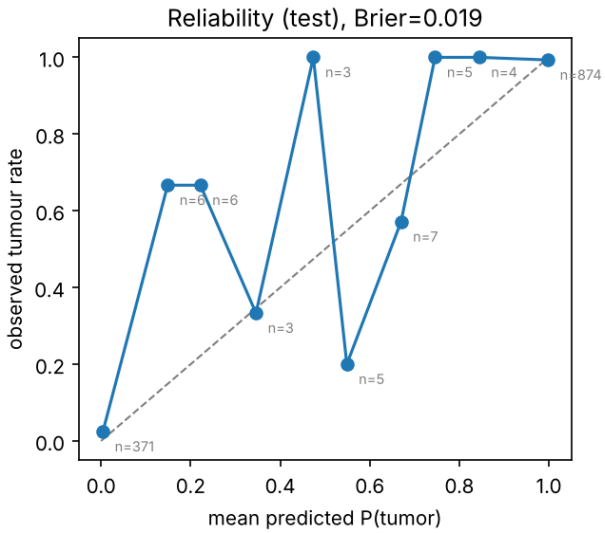


Figure 5 — Reliability diagram on the test set (Brier 0.019); n = scans per bin. The two heavily populated extreme bins lie on the diagonal, so the probabilities are usable as a triage priority; the sparsely populated middle bins are too small to judge.

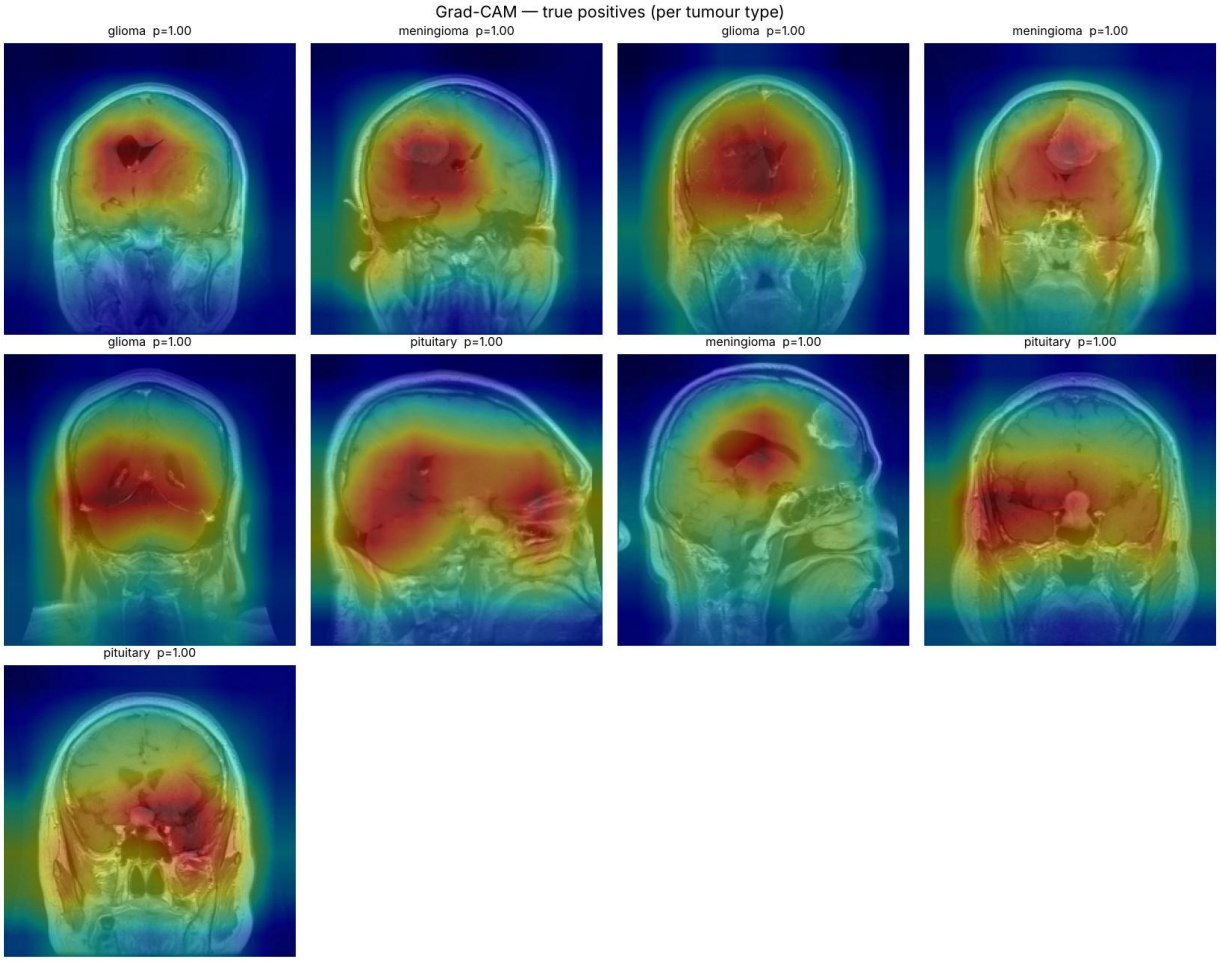


Figure 6 — Grad-CAM (layer top_conv) for correctly detected tumours of each type. Activation concentrates on the lesion and its surroundings rather than on skull or background.

9. Error Analysis

The test set contains 3 false negatives (meningioma 3) and 48 false positives. Every false negative is shown below with its score and Grad-CAM.

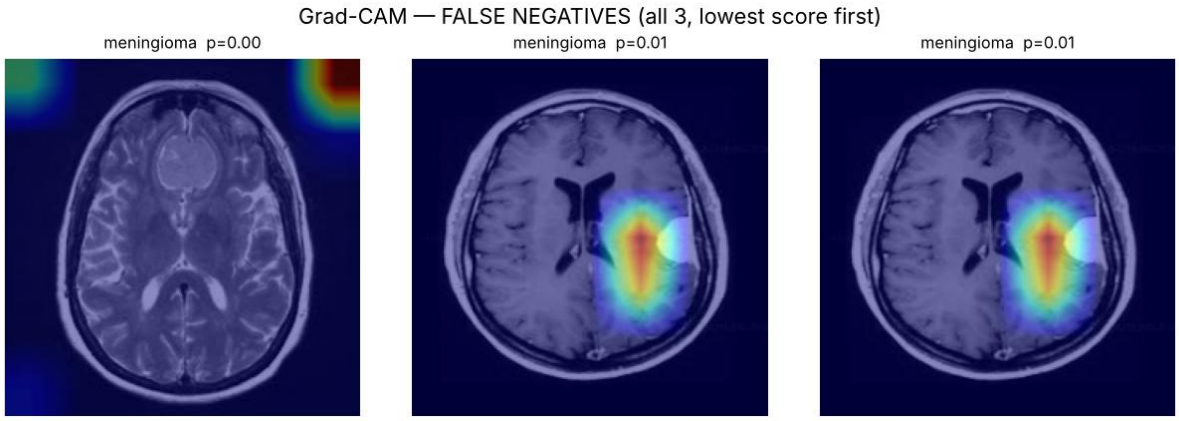


Figure 7 — All false negatives on the test set (lowest score first).

- **False negatives:** all 3 missed cases are meningioma 3 with P(tumour) ≤ 0.007 — confident misses, not borderline scores, so no threshold adjustment would recover them. In the Grad-CAM the network does attend to the lesion region in some of these slices yet still scores them as normal: the lesions are small, iso-intense with surrounding tissue and, in one case, the slice itself is atypical (the activation falls on the image corners, a sign the scan is out of the training distribution). Two of the three are near-identical slices of the same study, so the misses amount to two distinct cases. Meningiomas are also the weakest class in Table 6, consistent with their often subtle, dura-adjacent appearance.
- **False positives:** 48 normal scans exceed τ; 13 of them score above 0.5, the rest lie in the low-score tail that the recall-first threshold deliberately includes. They are typically scans with strong local contrast (ventricles, flow artefacts, bright white-matter lesions of non-tumour origin). Because the output is a triage priority rather than a diagnosis, these cost radiologist time but not patient safety.
- **Improvement plan:** (i) higher-resolution backbone (EfficientNet-B3 at 300 px, the GPU default of the notebook); (ii) test-time augmentation and a B0+B3 ensemble to reduce score variance near τ; (iii) CLAHE / intensity standardisation to reduce scanner contrast variability; (iv) collecting more small-lesion examples, which the per-type table identifies as the weakest group.

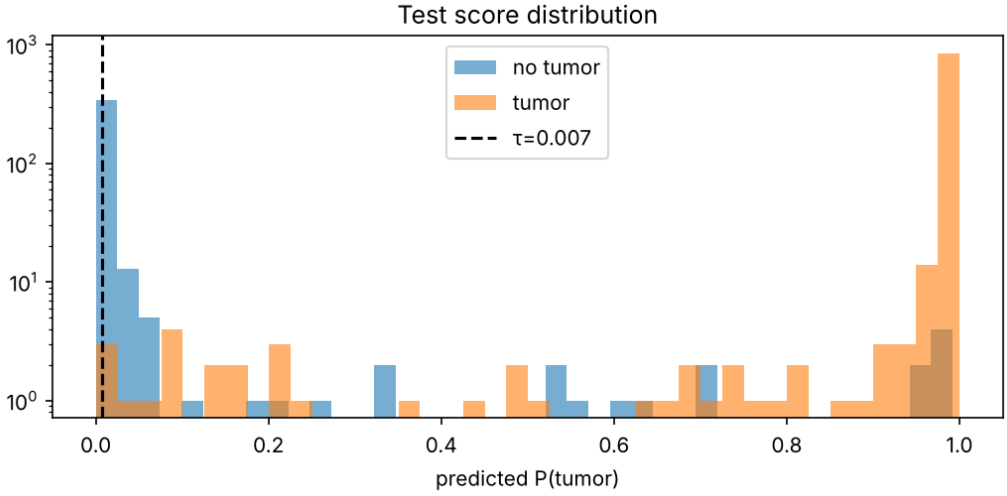


Figure 8 — Distribution of predicted P(tumour) on the test set (log scale) with τ. The two classes are almost completely separated; the false positives spread over the low-to-mid range that the recall-first threshold includes, while the 3 missed tumours sit inside the low-score mass of the normal scans.

10. Reflection, Limitations & Future Work

This project applies the Week 1–5 material end to end: formulating the task in DHAL terms (Weeks 1–2); recognising the mismatch between the training loss and the deployment criterion and resolving it with constrained threshold selection (Week 3); controlling hypothesis capacity through staged fine-tuning, dropout and augmentation while watching for overfitting on the learning curves (Week 4); and tuning the optimisation schedule — learning rates, early stopping and LR decay — with a threshold-free monitoring metric (Week 5).

Corrections made to the first version of this work

An earlier version of this pipeline reported strong metrics that did not hold up. Re-examining the notebook revealed four defects, all corrected here, and the comparison is instructive for the course themes:

1. Images were rescaled to [0,1] before an EfficientNet that normalises 0–255 input internally, crippling the pretrained features (frozen-backbone AUC-ROC 0.82 vs 0.98 after the fix).
2. tf.data caching was applied after augmentation, so the "augmentation" was a single fixed draw repeated every epoch.
3. The precision floor (0.60) was below the positive prior (0.71), so the recall-maximising rule selected $\tau = 0$ — a classifier that labels every scan as a tumour (specificity 0) while appearing to have "recall 1.0".
4. Model selection and early stopping monitored recall, which rewards exactly that degenerate solution; they now monitor AUC-PR. The random re-split also ignored duplicate images across the official splits.

Limitations

- The documented run uses EfficientNetB0 at 224×224 on CPU; the notebook defaults to EfficientNet-B3 at 300 px on a GPU, which should improve small-lesion sensitivity further.
- Single-slice 2D classification ignores volumetric context; the dataset mixes scanners, sequences and planes without metadata, so domain shift to a specific hospital is untested.
- Class labels are image-level; no lesion masks exist to verify Grad-CAM localisation quantitatively.

Future work

- 3D or multi-slice models with segmentation supervision for clinically verifiable localisation; prospective evaluation of the triage workflow (radiologist time saved per false alarm); calibration monitoring under distribution shift.

11. Conclusion

A two-stage fine-tuned EfficientNetB0 with a constraint-aware operating point detects 99.7% of tumour scans on a leak-free held-out test set while keeping specificity at 87.4% and AUC-ROC at 0.998. The central lesson is methodological: the criterion must be designed so that the trivial solution is infeasible, and the model must be selected on a metric that cannot be gamed by that solution — otherwise impressive-looking recall is meaningless. All reported numbers are reproducible from the notebook and its metrics.json.

12. Generative AI Use Statement

Generative AI (Anthropic Claude) was used as a coding and writing assistant: reviewing the first version of the notebook and identifying the defects listed in Section 10, restructuring the data pipeline and evaluation code, and drafting and editing the prose of this report. All experiments were executed by the

student from the submitted notebook, and every number, table and figure in this report is taken directly from that notebook's output files. The student is responsible for the final content.

13. References

- Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML 2019. arXiv:1905.11946.
- Nickparvar, M. (2021). Brain Tumor MRI Dataset. Kaggle. https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset
- Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV 2017.
- Saito, T., & Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. PLOS ONE 10(3).
- Deng, J., et al. (2009). ImageNet: A large-scale hierarchical image database. CVPR 2009.
- Chollet, F., et al. Keras Applications — EfficientNet. https://keras.io/api/applications/efficientnet/